ELEC5516
ELECTRICAL AND OPTICAL SENSOR DESIGN
LAB 5 Integrated Photonics Design – Microring Resonator

1 Objective

This laboratory aims to design an integrated photonics sensor application aided by Computer-aided Design (CAD) Software – ANSYS Lumericals. Note that this lab can be considered an open-ended design project.

The merit of the work is evaluated by the competence in designing an engineering solution aided by CAD. The outcome objectives are to

a. Apply knowledge of the concepts and the principles (that you have studied in ELEC5516) in the optical part to the design of photonic sensor applications.
b. Construct simulation models to represent the designs by demonstrating the effective use of computer-based design tools of analysis, visualisation, and modelling.
c. Present reports communicating technical and often complex material using clear and concise language

2 Introduction

Silicon-on-Instulator (SOI) can be fabricated with standard complementary metal–oxide–semiconductor (CMOS) processing, such as UV lithography. This enables mass fabrication. Besides, SOI allows high refractive index contrast for the nanophotonic circuits at the micron and submicron scale.

Optical Ring Resonator topology has found many applications in optical communications and signal filtering. By implementing Ring Resonator Structure based on the platform of SOI, we can design microring structures (ring resonator as the building block) and develop applications suitable for integrated optical sensors, including the field of biosensing. In this lab, we are designing integrated optical sensors with CAD.

3 CAD Software

Ansys Lumerical suite is a powerful CAD software that provides multiphysics simulation tools, including Finite-difference Eigenmode Solver (for waveguide design), Finite Difference Time

Domain Solver (2D/3D FDTD suited for nanophotonics) and Scattering Matrix Algorithm (1D travelling wave for circuit design and Simulation).

Those tools enable the design of photonic components, circuits, and systems in one software framework, as shown in Figure 1. In particular, device and system-level models can work together seamlessly, interacting with optical and electrical effects. In this lab, we use Ansys Lumerical's suite to complete our experiments.

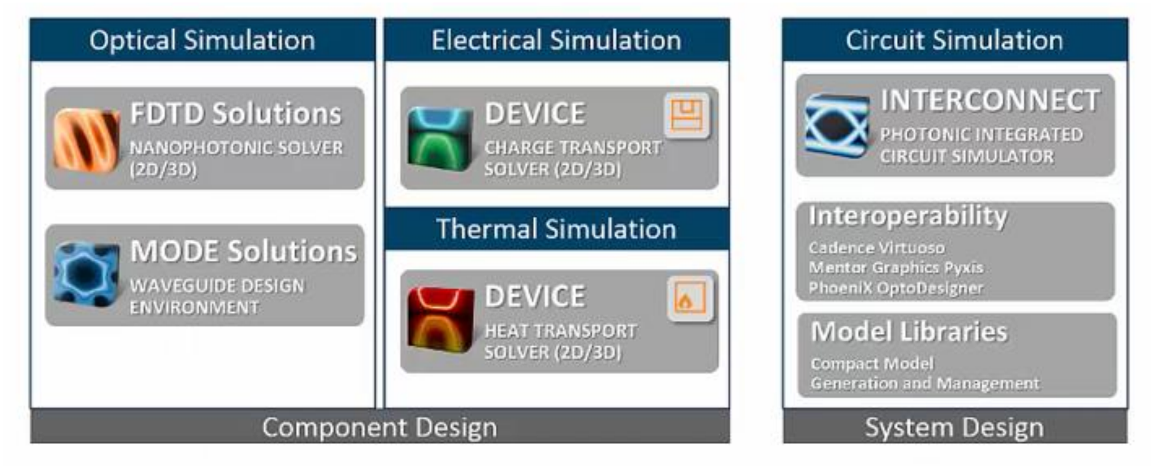


Figure 1: Ansys Lumerical CAD.

In this lab, we start with modelling a ring resonator structure on the system level with INTERCONNECT. Then we deep dive into the device parameter characterisation on the device level with a physics-based simulation tool such as MODE. Note that fully developing a realistic system requires going back and forth between device-level and system-level simulations.

4 Ring Resonator Structure Basics

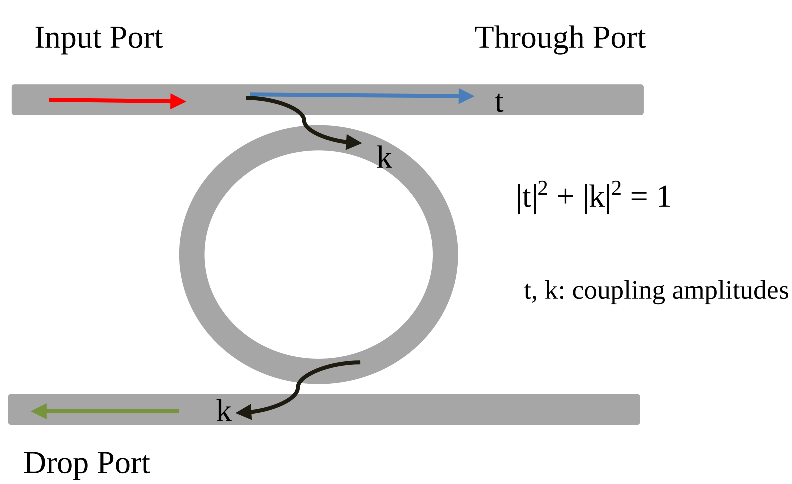


Figure 2: Illustration of a single-ring resonator structure.

In its basic form, a single-ring resonator is comprised of two straight waveguides and a ring waveguide, as shown in Figure 2. Its functional terminals include Input Port, Through Port and Drop Port. It works based on the principle of constructive and destructive interference.

The input light couples into the ring waveguide. After propagating around it, some light couples back to the straight waveguide and interacts with the incident light. At resonance (i.e. optical propagation length of the Ring is equivalent to an integral number of the wavelengths), completely destructive interference occurs at the Through Port, resulting in no transmitted light.

On the other side, a strong optical field can build up in the ring structure at the resonant wavelength due to constructive interference and couples to the Drop port resulting in power transfer at the Drop port at the resonant wavelength.

A series of the resonant wavelengths (or frequency), as shown in Figure 2, are governed by the radius of the ring resonator (r) and the effective index of the waveguide (n_{eff}), which can be expressed as

$$N\,\lambda_{res} = 2\pi\,\mathrm{r}\,n_{eff} \tag{1}$$

where N is an integer number, and the *perimeter* of the Ring is $L = 2\pi\,\mathrm{r}$.

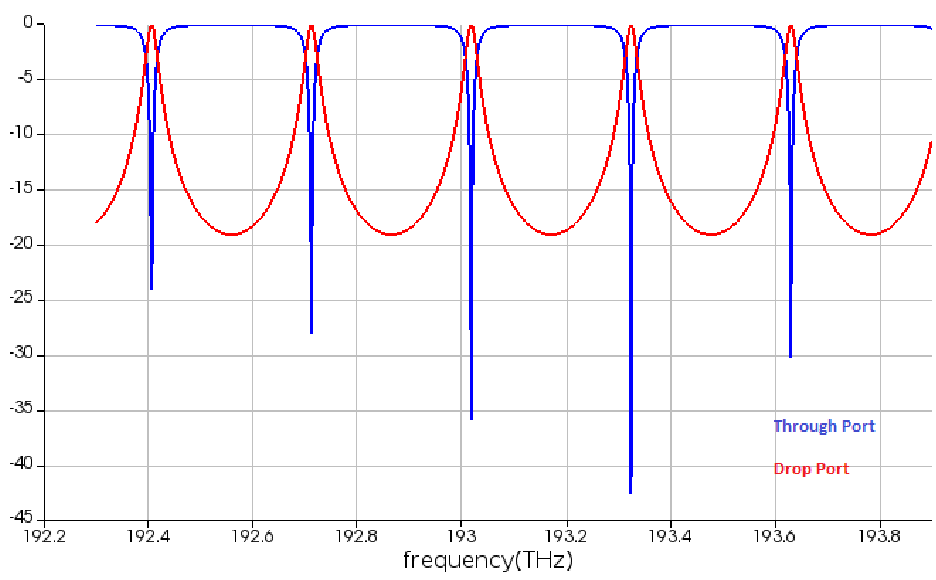


Figure 2: Transfer Functions at Through port and Drop Port.

Key specifications of a ring resonator structure include Free Spectral Range (FSR), Quality Factor (QF) and Finesse.

- ***FSR*** is the difference between two optical frequencies (or wavelengths) of successive reflected (or transmitted) maxima (or minima). It is defined as:

$$FSR = \frac{c}{n_g(2\pi\,\mathrm{r})} \quad \text{in frequency} \tag{2a}$$

$$FSR = \frac{\lambda_{res}^2}{n_g(2\pi\,\mathrm{r})} \quad \text{in wavelength} \tag{2b}$$

  c is the speed of light. Note that in Eqn (2), we use the group index (n_g) instead of the effective index to account for its dependence on the optical frequency. The effective refractive index is related to the group index by

$$n_g = n_{eff} - \lambda_{res}\frac{d\,n_{eff}}{d\lambda} \tag{2c}$$

- ***QF*** is defined as the ratio between the operating frequency to the Full-width half max of the transfer function at the Drop Port. When loss is ignored, it can be **approximated** by

$$QF = \frac{n_g(2\pi\,\mathrm{r})\pi}{\lambda_{res}}\frac{|t|}{1-|t|^2} \tag{3}$$

- ***Finesse*** is defined as the ratio between FSR to Full-width half max of the transfer function at the Drop Port. When loss is ignored, it can be **approximated** by

$$Finesse = \frac{\pi|t|}{1-|t|^2} \tag{4}$$

5 Photonics Circuit-Level Modeling

Lumerical INTERCONNECT is photonics circuit-level modelling software. Open the model "*ring_resonator student.icp*". Spend some time to know the software interface as shown in Figure 3. These modules/components and connections required for this experiment have been placed in the Schematic Editor window. Note that we can also locate these modules from **Element Library**.

Click each block, and we can view its primary variables in its **Property View** window and pay attention to these configurable parameters under the **Standard** section and **Waveguide** section, where we can specify the system parameters for the simulations. We can right-click on the module and find Help which directs us to the module's online documentation.

As shown in Figure 4, a single-ring resonator may be modelled by optical circuit components consisting of straight waveguides and waveguide couplers. Investigate the structure in Figure 4 by referring to Figure 2.

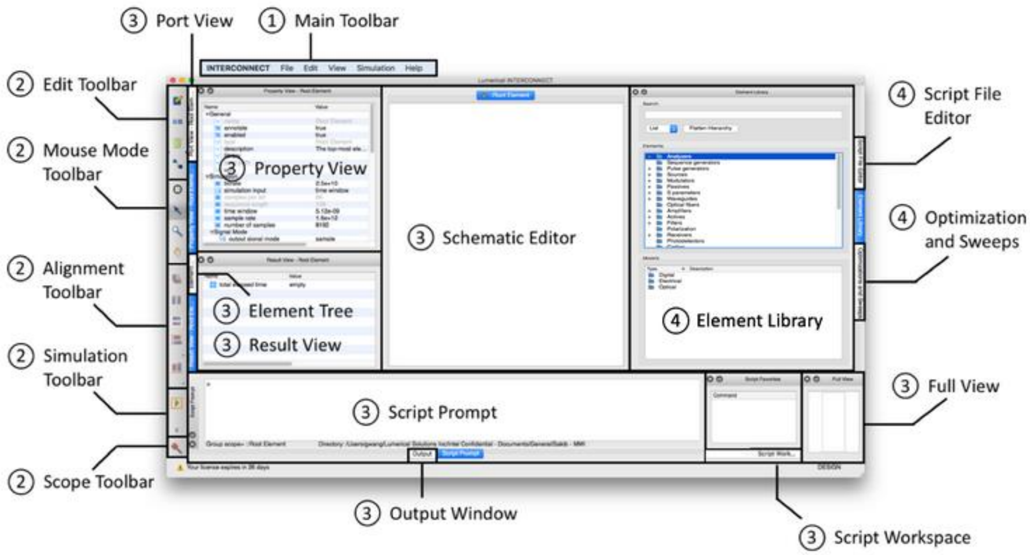


Figure 3 Lumerical INTERCONNECT Interface (Source: Lumerical INTERCONNECT)

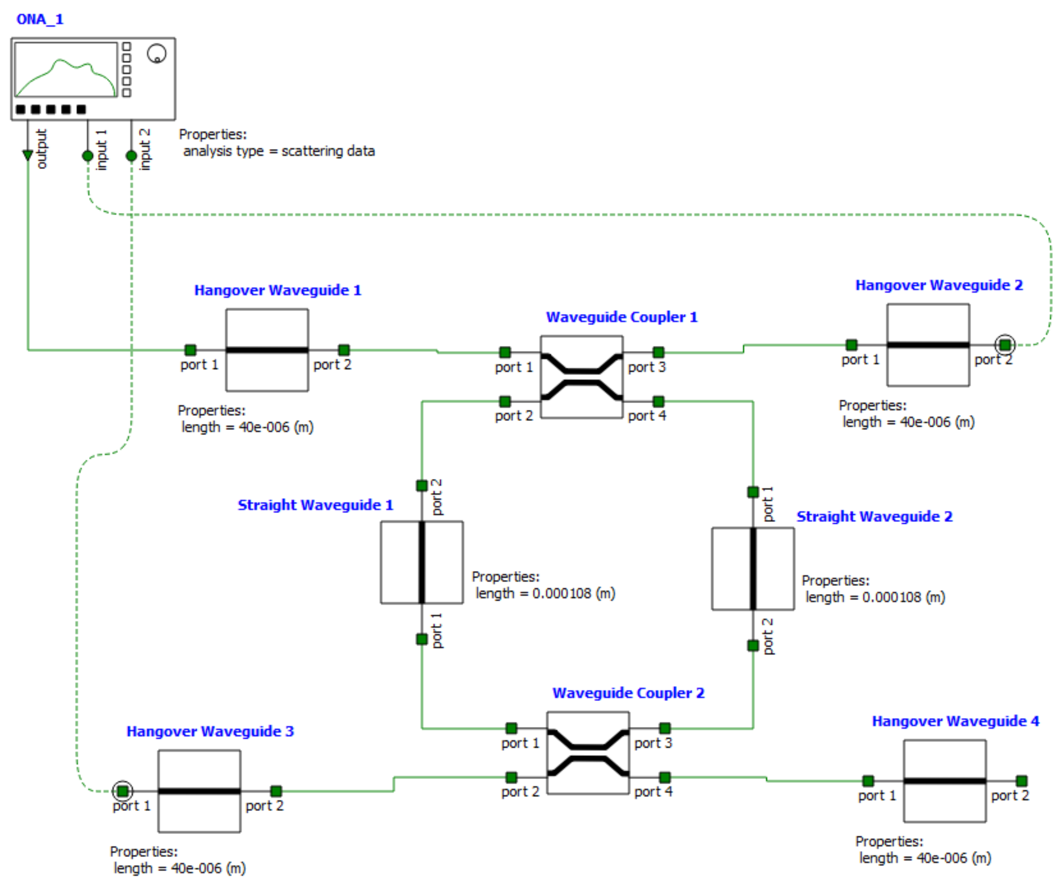
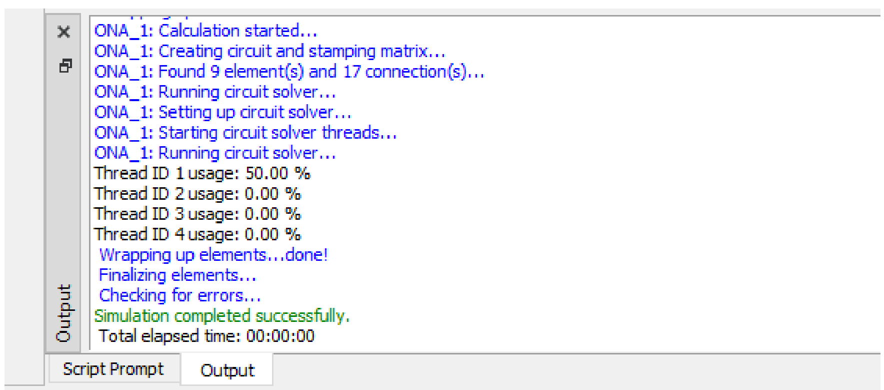


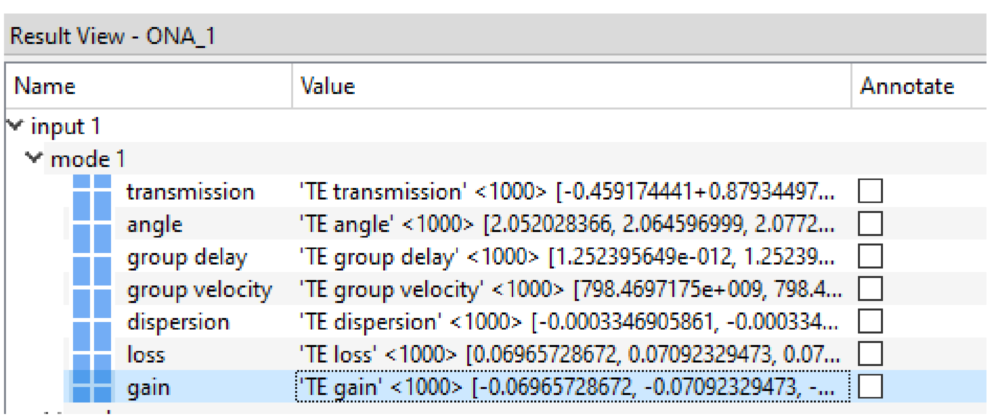
Figure 4 Circuit Model of Ring Resonator

Experiment Procedures

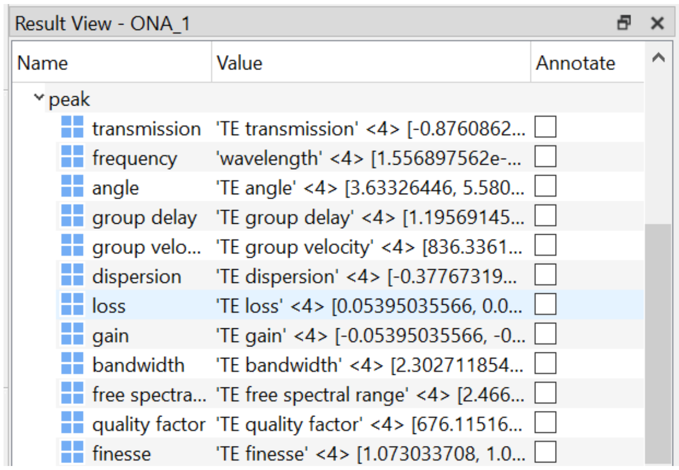
1) Assuming single-mode operation, TE (i.e. Mode 1) as the fundamental mode is analysed in this Simulation.
2) By referring to Eqn 1 to 4, we know that the key **system parameters** that determine the behaviours of the ring structure include *group index* (ng), the *radius* of the Ring (*r*) or the length of the *ring perimeter* (L=2*pi*r), and *power coupling ratio* ($|t|^2$). **Locate and calculate** these parameters from the simulation model. Note that the coupling coefficient of the simulation model - coupler is the power ratio defined as $|k|^2$, as shown in Figure 2.
3) Optical Network Analyzer (**ONA**) is a virtual instrument to measure the frequency response of the photonic circuit under test by impulse response analysis. Understand the parameters under the Property View used in this simulation model.
4) Run the Simulation by pressing the button, and observe the **Output** terminal to monitor the progress of the simulation calculation. Ensure that there is no error message.



5) To observe the transfer function at the Through port, click the ONA_1. Note that only Mode 1 (i.e. TE mode) is to be investigated. In the Result View window, select input 1 -> mode 1 -> and then right-click the **gain** variable to visualise. **Record and describe the figure.**



6) Similarly, to observe the transfer function at the Drop port, click the OVN_1. In the Result View window, select input 2 -> mode 1 -> and then right-click the **gain** variable to visualise. **Record and describe the figure.**
7) **Verify and comment** on the simulation results with the theory, i.e. Eqn 2- 4. Note that we may obtain FSR, QF and Finesse by directly reading off points from the figures. Alternatively, in the Result View window, we can get those parameters under **peak** directly



8) **(Open-end question) modify** one of the parameters: one roundtrip length, coupling coefficient or group index, and **explain qualitatively or/and quantitatively** how this may affect the system's behaviours. Fundamentals of ring-based sensor applications are based on measuring the scanning spectrum as a result of the change in one of these parameters due to external disturbance. For example, in some biosensing applications, when the test solution is flown over the ring resonator, analyte molecules would specifically bind to the immobilised receptors in the evanescent field of the resonator waveguide. This increases the effective roundtrip length (or changes the refractive index) of the resonator.

In this experiment, we may appreciate the fact that the characteristics of the ring resonator are determined by its system parameters (i.e. circuit parameters). In the next experiment, we will investigate the device parameters based on **Silicon on Insulator (SOI)** that derive the system parameters. When we fabricate the integrated photonics circuits, it is the device parameters that are required to be processed.

6 Photonics Device-Level Modelling

6.1 SOI Platform

The SOI platform consists of a Silicon core (Si, **n~3.47**) and a Silicon Oxide (SiO_2, **n~1.44**) cladding bottom. The typical Silicon waveguide has its cross-section: a width of 450nm and a height of 220nm. As observed, the index difference (**~2**) between the core and cladding is significant and indicates strong confinement of light (for TE-polarisation) within the waveguide. This makes SOI an ideal platform for integrated photonics.

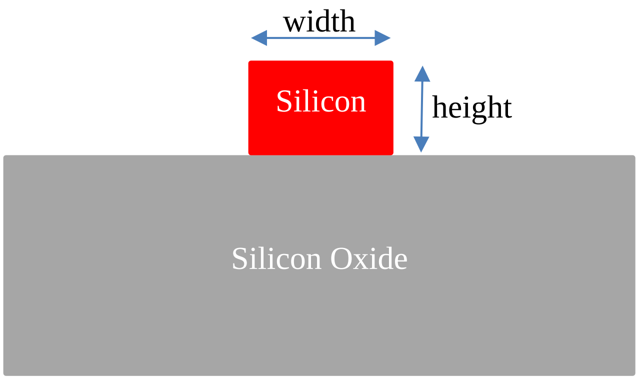


Figure 5 SOI waveguide

Lumerical Mode Solver is photonics device-level modelling software. Open the model "*ring_resonator student_fde.lms*". A preconfigured ring resonator model is shown in Figure 6. The model consists of one rectangle structure (SiO_2), and one ring resonator structure (Si). It also has preconfigured two simulation solvers: i.e. FDE and varFDTD. Spend some time to know the software interface.

FDE solver solves the vectorial Maxwell's equations at a **single frequency** based on material and geometry specified by the user. The varFDTD solver simulates **the propagation** of light in SOI, by collapsing a 3D geometry into a 2D set of effective indices. As it is a time-domain method, one simulation results in broadband results. In this experiment, we will explore both FDE and varFDTD.

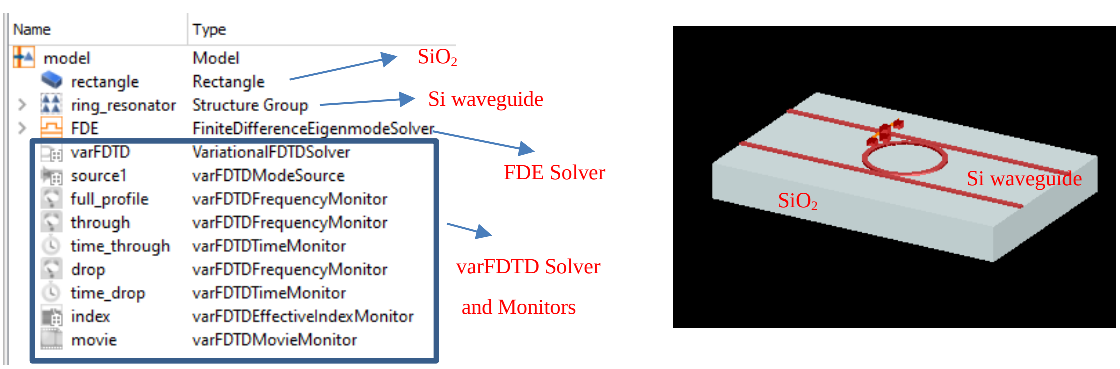


Figure 6 Single-ring device model

6.2 Design Question

Design Goal: design a ring resonator structure with an FSR of 26.2nm near 1550nm. It is based on the SOI platform of **400nm by 200nm** waveguide cross-section. The quality factor of the device is aimed to be 1800 at the Drop port, and Finesse is close to 26 at the Drop port.

Design equations are provided for your reference below.

- The radius of the Ring can be derived from the below equation, where L is the length of the Ring.

$$FSR = \frac{\lambda_{res}^2}{L\, n_g} \tag{5}$$

- The coupling amplitude (t) and the coupling length (L_c) of the Ring can be derived from the below equation.

$$|t| = \sqrt{1 + \left(\frac{n_g L\pi}{2Q\lambda}\right)^2} - \frac{n_g L\pi}{2Q\lambda} \tag{6a}$$

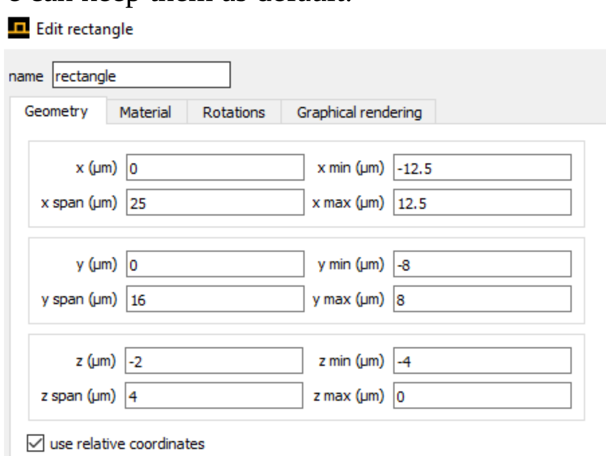
$$L_c = \frac{\lambda_{res}}{\pi\Delta n}\ sin^{-1}(\,|k|) \tag{6b}$$

where t and k are related to the coupling coefficients defined in Figure 2. Δn is the difference in the effective indexes of the first two coupling modes.

6.3 Experiment Procedures for FDE

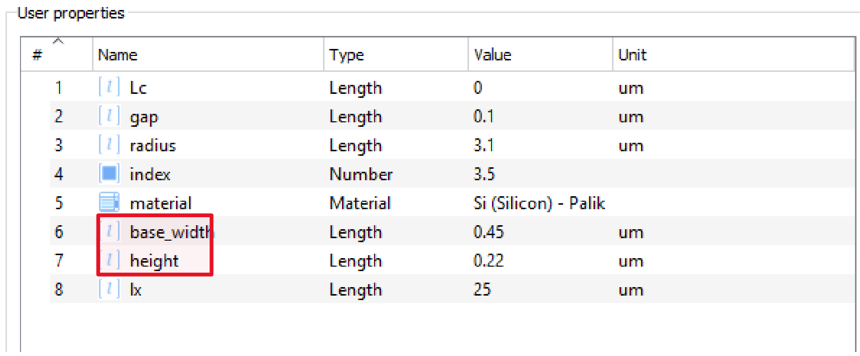
Experiment Procedures for FDE are provided below.

1) Open the model "*ring_resonator student_fde.lms*"
2) Right-click the **rectangle** object in the Object Tree, and click Edit Object. Then explore Geometry and Material Tabs. The parameters defined here affect the structure layout in the Simulation model. We can keep them as default.



3) Right-click **ring_ressonator** in the Object Tree, and click Edit Object. Then explore Geometry and Material Tabs. Spend some time exploring the User properties. The cross-

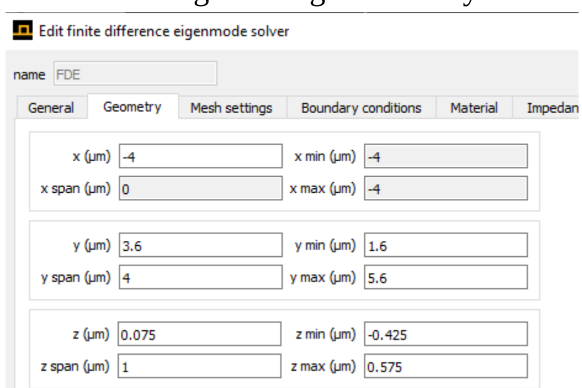
section of the waveguide is defined by base_width and height that **are entered according to the design goals**. We may vary some of the other properties to observe the object displayed in the Simulation Window. For the time being, restore it to the default and only change the base_width and height.



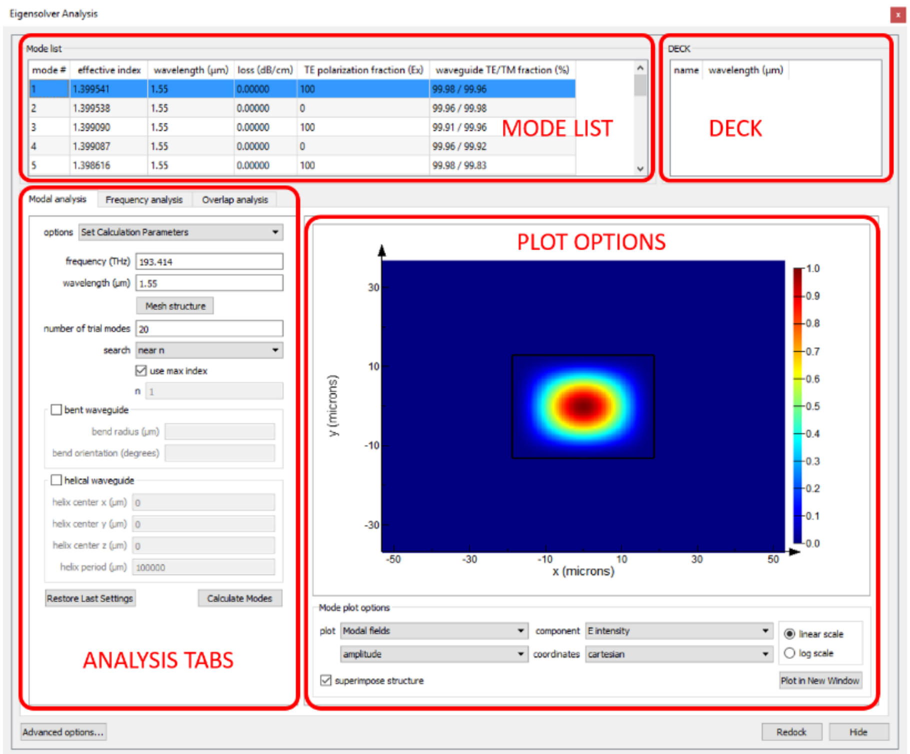
User properties

#	Name	Type	Value	Unit
1	Lc	Length	0	um
2	gap	Length	0.1	um
3	radius	Length	3.1	um
4	index	Number	3.5	
5	material	Material	Si (Silicon) - Palik	
6	base_width	Length	0.45	um
7	height	Length	0.22	um
8	lx	Length	25	um

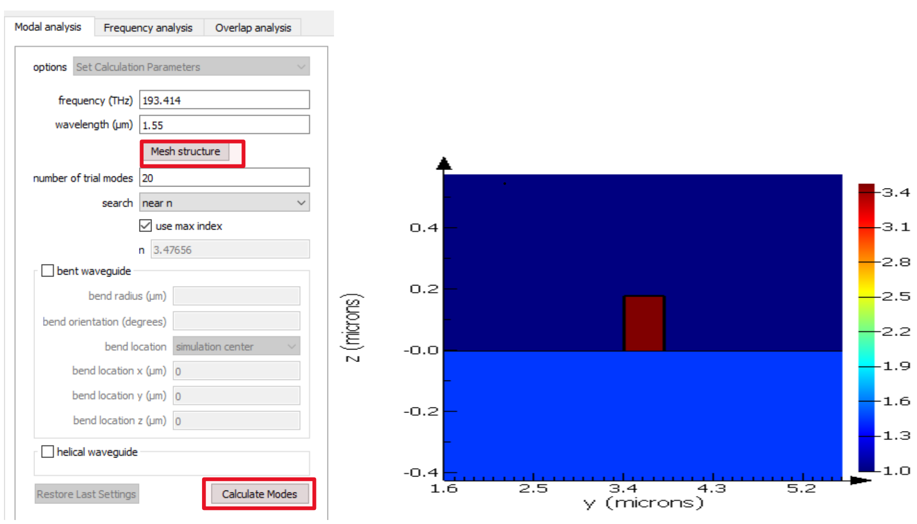
4) Right-click **FDE** and click Edit Object. The 2D solver region is defined in the Geometry tap. And the cross-section of the straight waveguide is fully contained within the region.



5) To obtain an effective index and group index of the modes, modal analysis is to be performed. Below are the analysis windows (Source: Lumerical Online article).



6) Click the **Mesh structure** button to observe the material property (i.e. refractive index) of the cross-section.

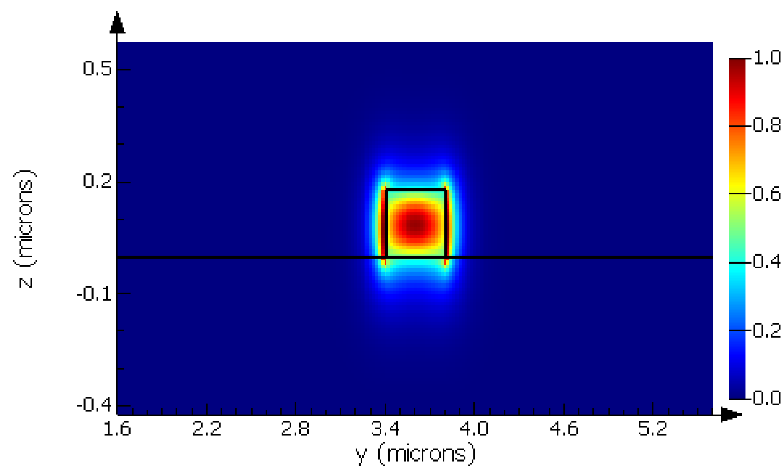


7) Then click the **Calculate Modes** button to obtain the modes for this waveguide. Twenty modes that are calculated appear in the Mode list; the first one is the fundamental TE mode.

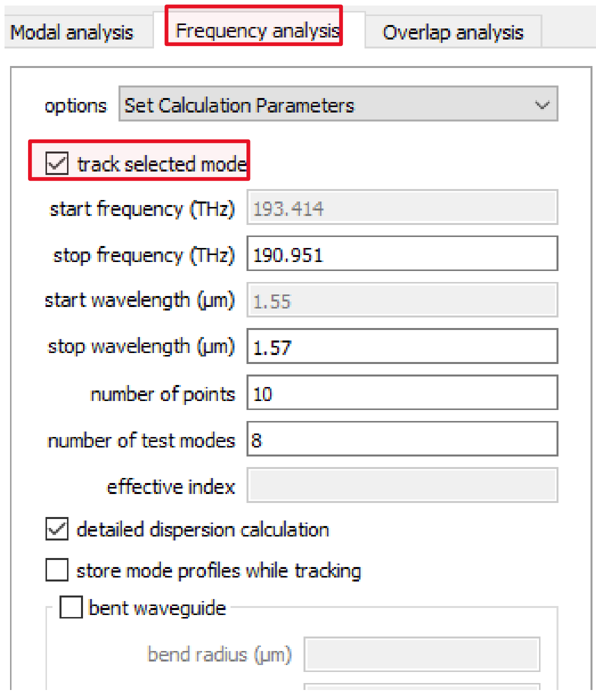
Mode list

mode #	effective index	wavelength (µm)	loss (dB/µm)	group index	TE po fract
1	1.912092	1.55	0.0000	4.625830	94
2	1.421551	1.55	0.0000	1.874259	6
3	1.221309	1.55	0.0000	1.571117	2
4	1.184178	1.55	0.0000	1.660046	2
5	1.036685	1.55	0.0000	1.889612	5
6	1.034419	1.55	0.0000	1.508818	35
7	0.9293884	1.55	0.0000	1.768717	100

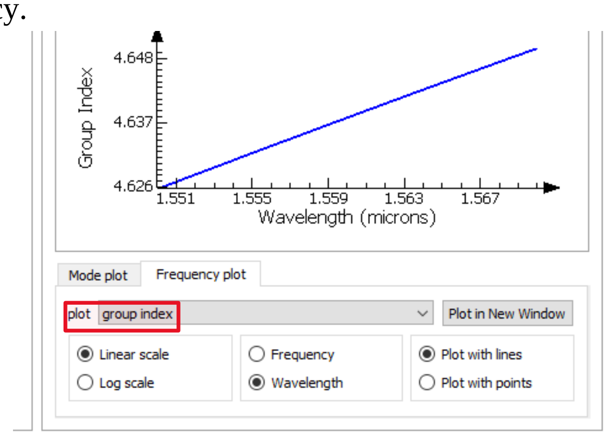
8) Record its **effective index** and **group index**. Also, record its 2D modal fields plot. It is important to note that the **E-field intensity** should be close to zero at the boundary of the solver region. Otherwise, consider increasing the size of the solver region.



9) **Frequency analysis** allows wavelength dependence characteristics to be investigated. Select the **first mode** from the mode list, tick the box of track selected mode, and perform Frequency Sweep (**up to 1600nm**).



10) Observe and record the **effective index** and **group index** on the dependence of the wavelength/frequency.



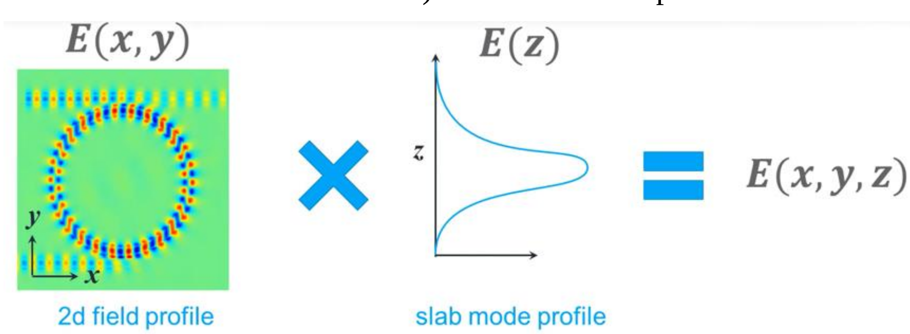
11) Since we have obtained the group index, use Eqn (5) to calculate the **total length** of the Ring or the radius based on the design goal.
12) Next, we calculate the **coupling length** based on Eqn (6). Δn can be calculated by the difference in the effective index of the symmetric and anti-symmetric coupled modes. To calculate the coupled modes, right-click **FDE** in the Object Tree and Edit its geometry property as below. Essentially, the FDE region contains both the cross-section of the straight waveguide and that of the Ring.

13) Click Calucate Modes, and the **first two modes** in the mode lists are the symmetric and anti-symmetric coupled modes required for the calculation. Update the radius (adjusted), and *Lc* in the User properties of the ring_resonator object accordingly.
14) So far, we have derived all the **physical device parameters** required to meet the design goal. Next, we will use **varFDTD** to verify the design.
15) **(Open-end question) Modify the Ring and straight Waveguide gap** (the most common gap is between 80nm to 150nm). How would it affect the other device parameters?
16) **(Open-end question) Modify the Ring's cross-section** (the most commonly used dimensions are width between 380 nm to 460nm and height between 180 nm to 240 nm). How would it affect the other device parameters?

7 Design Verification with varFDTD

The **finite-difference time-domain** (FDTD) is another method to solve Maxwell's equations numerically. It has found applications with dimension size on the order of wavelength. As a time-domain technique, one Simulation can give broadband results by Fourier Transform, and besides it can also emulate the time-step propagation of optical waves. Those two features make the technique attractive in analysing the optical gain spectrum of the micro-ring resonator structure.

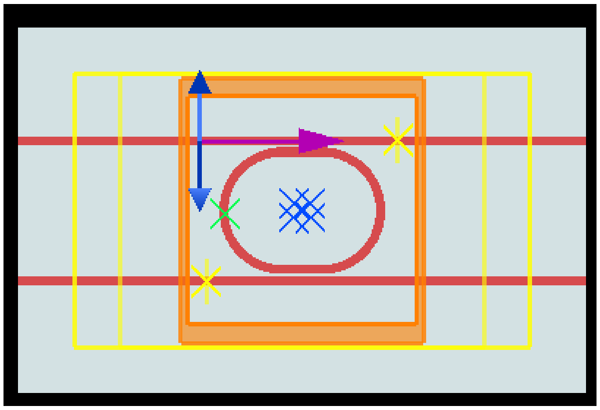
Variational FDTD (**varFDTD**) is an extension of FDTD by collapsing a 3D geometry to 2D. It is achieved by expanding a 3D mode profile into a 2D field profile (based on 2D effective material indexes derived from the 3D material indexes) and a slab mode profile.



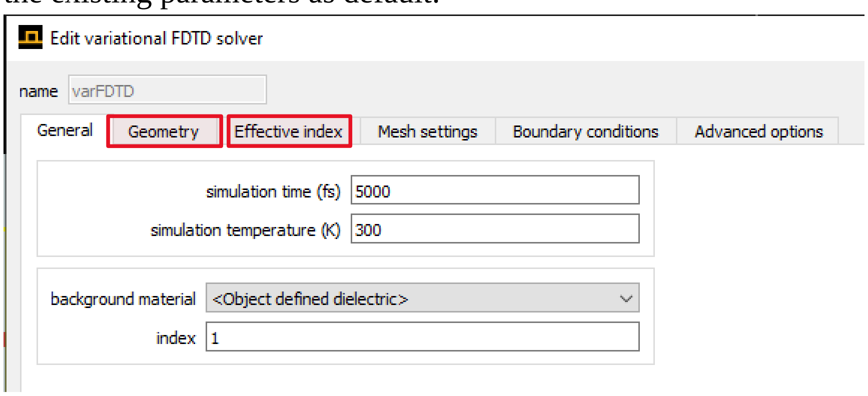
This extension aims to significantly reduce the computation resources required to accurately simulate nano-structures such as ring resonator structures in this lab.

In this Section, we use varFDTD to verify the design from the previous sections. **Experiment Procedures for this technique are provided below.**

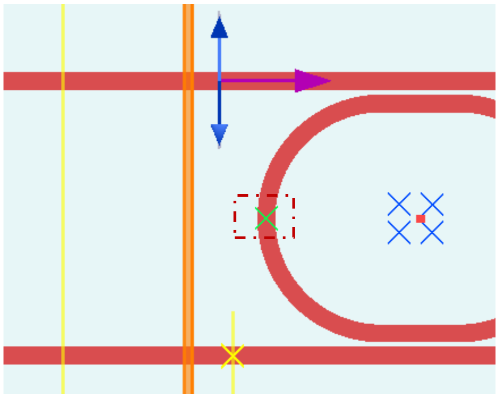
1) Open the model "*ring_resonator student_varfdtd.lms*". Enter the ring resonator parameters that meet the design goal which you have obtained from the previous Section.



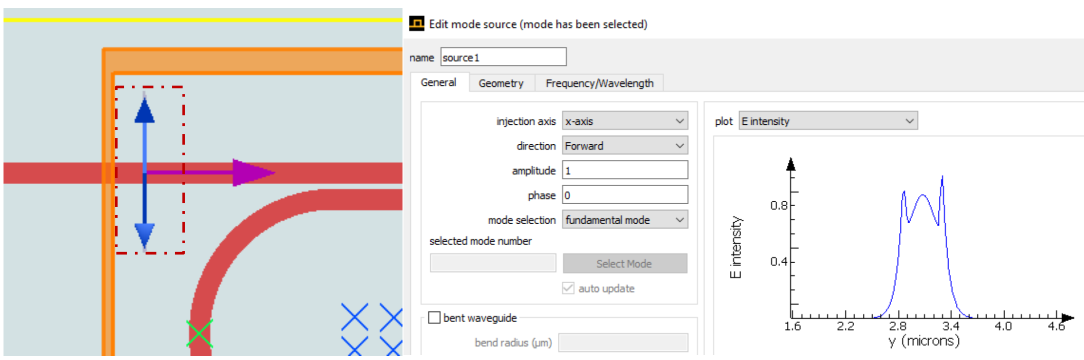
2) Explore the parameters of the varFDTD solver. The Geometry tab allows us to define the solver region (3D box shape). varFDTD uses **the effective index method** to collapse the 3D geometry into a 2D representation, which can be edited in the Effective index tab. Use the existing parameters as default.



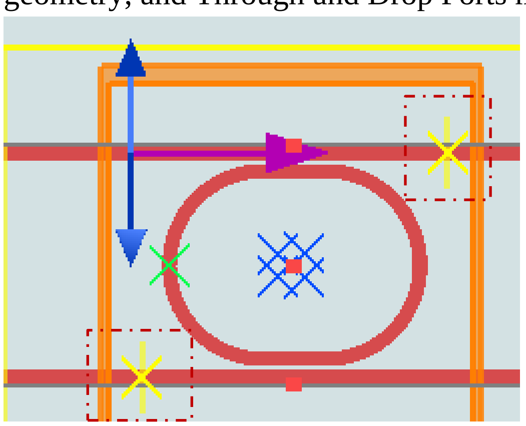
3) A **green cross** in the graphical user interface sets the location of the **core slab mode** for the model. Ensure that it overlaps with the Silicon portion of the ring resonator. Its position can be varied by dragging it directly or specifying the position under the Geometry tab.



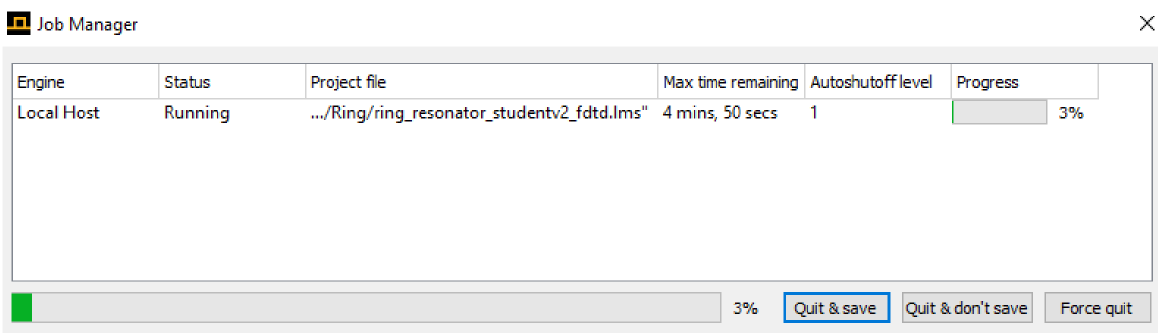
4) Explore the parameters of **varFDTDModeSource**. The mode source injects a guided mode (pulse) into the simulation waveguide. Use the existing parameters as default. And ensure that its position is at the centre of the input port.



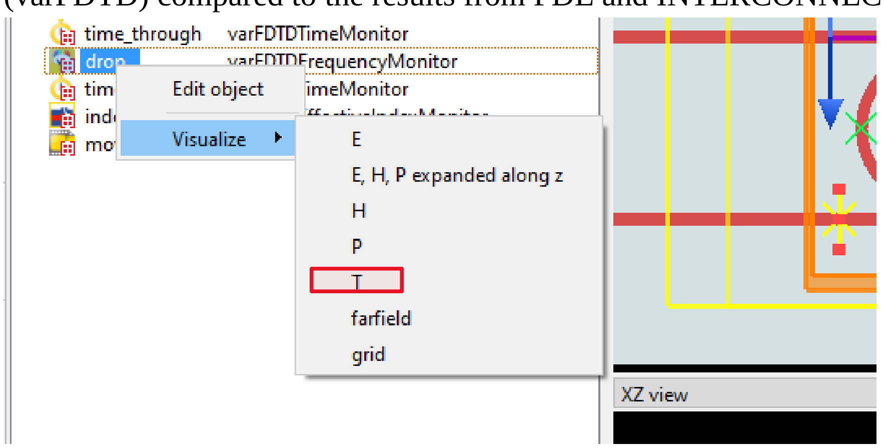
5) Explore the parameters of **varFDTDFrequcneyMonitor** at Full Profile, Through and Drop Ports. Frequency-domain field monitors collect the field profile in the frequency domain from simulation results across the spatial region. The Full Profile monitor has 2D geometry, and Through and Drop Ports monitors have 1D geometry.

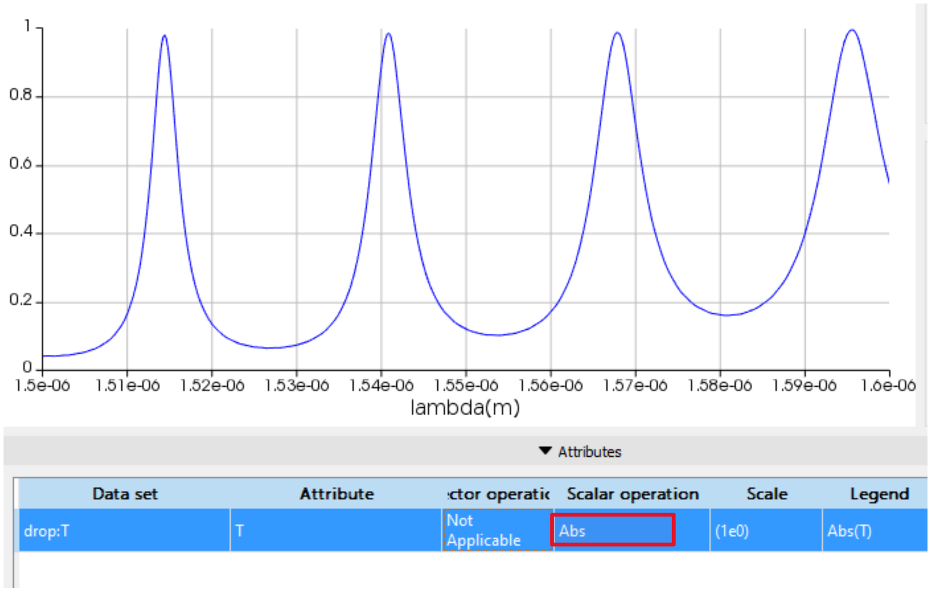


6) Explore the parameters of **varFDTDTimeMonitor** at Through and Drop Ports. These monitors consist of measurement points that capture time-domain information for field components at that point location over the simulation period.
7) Explore the parameters of **varFDTDEffectiveIndexMonitor,** which has 2D geometry covering the entire ring resonator structure. It captures the effective index values as a function of wavelength.
8) Explore the parameters of **varFDTDMovieMonitor**, which has 2D geometry covering the entire ring resonator structure. It records an animation of desired field component over the region during the simulation period.
9) Press on the **RUN button** to start the Simulation. It may take over 5 minutes to complete the Simulation, depending on the computer.



10) **(Open-end question)** After the Simulation, **inspect, record and explain** the results from those Monitors by **visualising them**. Note that the movie file (.m1v) is saved to the same directory as the simulation file.
11) The key feature of the ring resonator is its FSR. It can be observed by visualising T (transmission) of **varFDTDFrequcneyMonitor** at the drop port. Measure its FSR (varFDTD) compared to the results from FDE and INTERCONNECT.





8 Discussion

Although **QF** and **Finnese** are important features of the ring resonator, in practice, FSR is the only fundamental indicator commonly used in photonic sensing applications. This is because the signal losses heavily influence QF and Finnese, and their values can be sensitively affected by any minor numerical errors (inherent) and standard fabrication variations.

In this lab, we modelled a ring resonator structure from the system and device levels. Each type of Simulation has its advantage and disadvantage, and sound design requires multiple interactions between those simulations at the different abstract levels.

Note that the object in this lab only concerns a single-ring structure, and we can imagine that advanced integrated photonics designs involving multiple rings with various device parameters at the device level can exert much more effort to complete, especially extending to full 3D FDTD. That is why Hign Performance Computing is much needed to tackle such problems. However, elegant design often starts with first understanding the design goals and requirements. And then, narrow down the range of device parameters using circuit simulators and FDE before passing them to the test of a full 3D FDTD.

Discussing with process engineers to understand the facilities' limitations in fabricating the photonics chips is also essential. Commonly, they can provide you with preferable cross sections of Si waveguide and the minimum achievable gap. Those inputs should provide feedback on your design. Besides, some standard fabrication variations/imperfections may also be provided so that we, as the designer, would sweep the device parameters to observe the range of outcomes resulting from our design (i.e. Monte Carlo analysis).